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Efficient Multi-Robot Task Planning with Large Language Models

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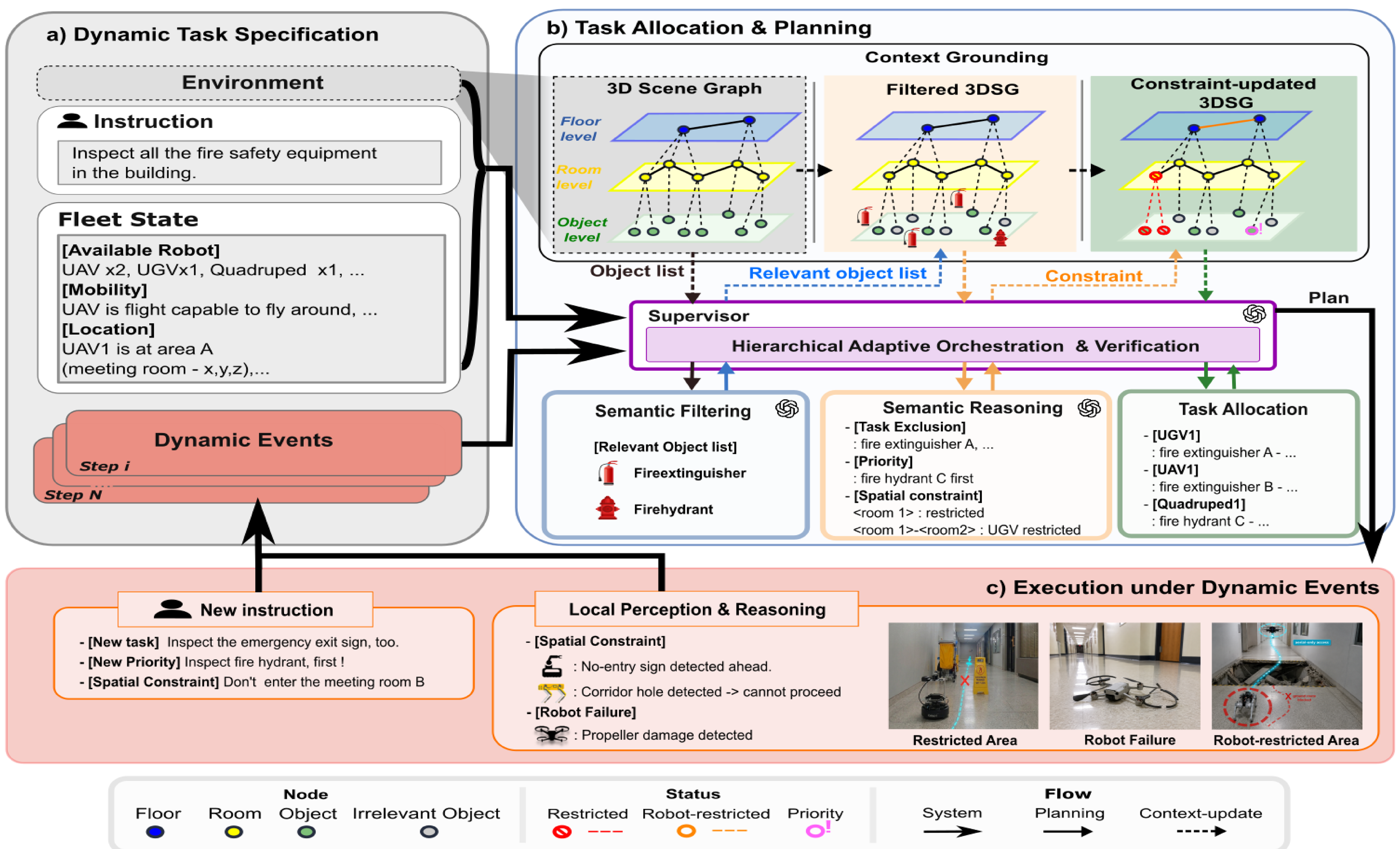
Introduction

- **Dynamic Multi-Robot Systems (MRS):** Must operate in unpredictable domains (e.g., search & rescue, logistics) and **adapt to operator instructions, environmental changes, and robot failures.**
- **Conventional Planners:** Efficiently optimize task allocation for **pre-defined problems**, but **lack the flexibility** to handle semantic and dynamic constraints.
- **LLM-based Planners:** Enable complex task decomposition and collaboration via natural language, but face **high planning costs** as tasks and constraints scale. [1-3]
- **The Challenge:** Achieving **efficient, constraint-aware task allocation and real-time replanning** remains difficult under LLM-based planning.

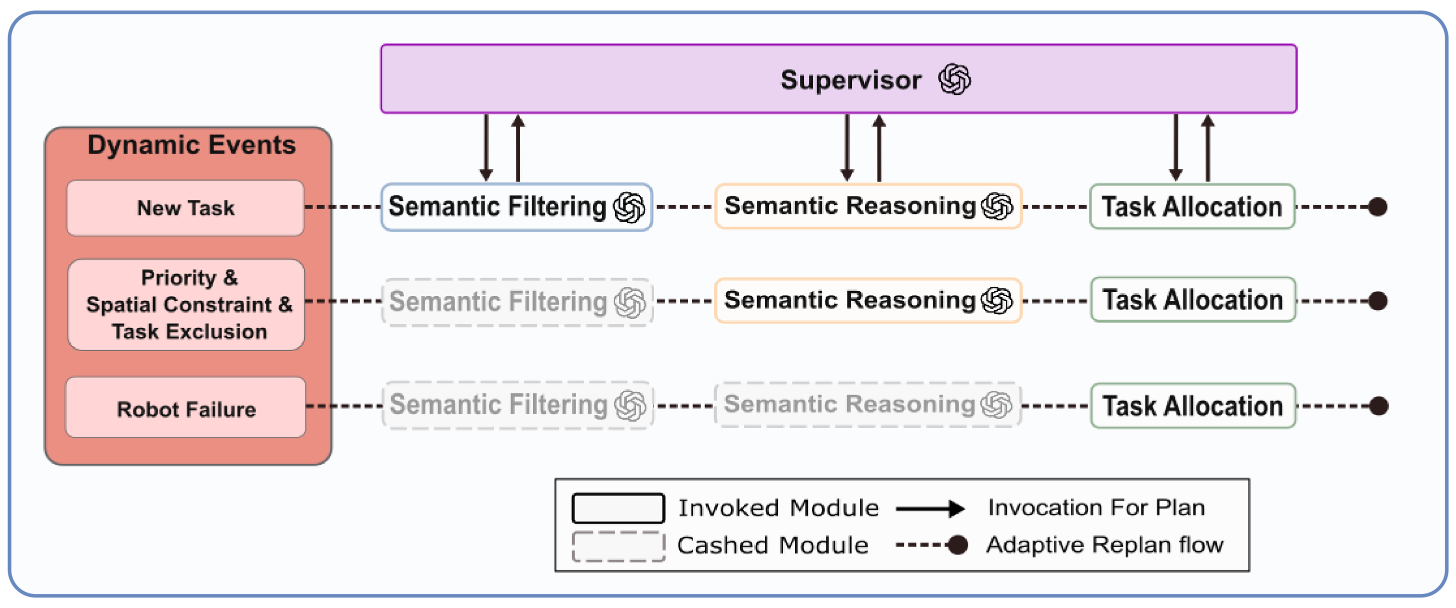
Overview



1. Framework

Contributions

- LLM-based dynamic MRTA framework for **efficient multi-task execution under complex constraints** and unpredictable environments.
- **Hierarchical adaptive LLM orchestration & 3D scene graph** representation to **reduce planning cost and latency**
- Validated in simulation and real-world experiments, demonstrating **high efficiency and near real-time replanning**.



2. Hierarchical Adaptive Orchestration

Simulations

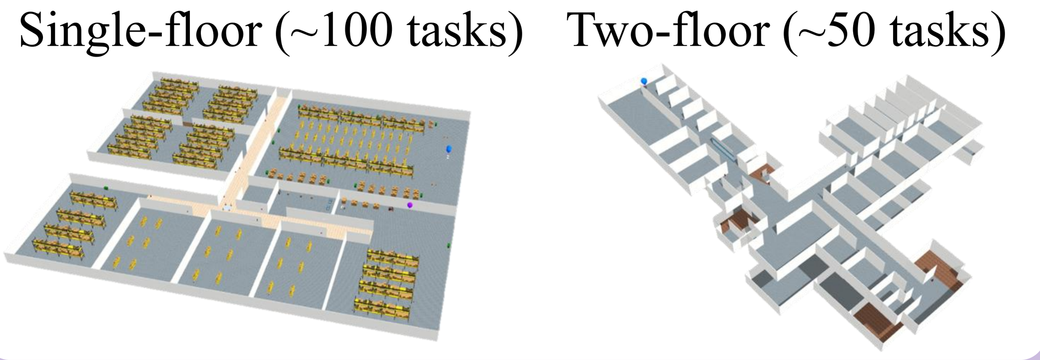
Setup

- **Task** : “Inspect all fire safety equipment in the building”
- **Robot Team** : 2 UGVs + 1 UAV
- **Constraints***
  - ➢ C1 **[Task Exclusion]**
  - ➢ C2 **[Task Priority]**
  - ➢ C3 **[Spatial Restriction]**
  - ➢ C4 **[Robot Failure]**
- **Scenarios**

1. **Initial planning**: evaluates initial planning performance for each constraint type.
2. **Online replanning**: evaluates replanning performance as randomly triggered constraints accumulate over time.
3. **Scalability**: increase the number of robots and tasks to analyze robustness and planning cost.

*For C1–C4, constraint expressions are varied by difficulty level and randomly sampled across trials.

Env. A Single-floor (~100 tasks) **Env. B** Two-floor (~50 tasks)



- **SR** : (Plan) Success Rate
- **TCR** : Task Completion Rate
- **CSR** : Constraint Satisfaction Rate
- **Time** : (Plan) computation time

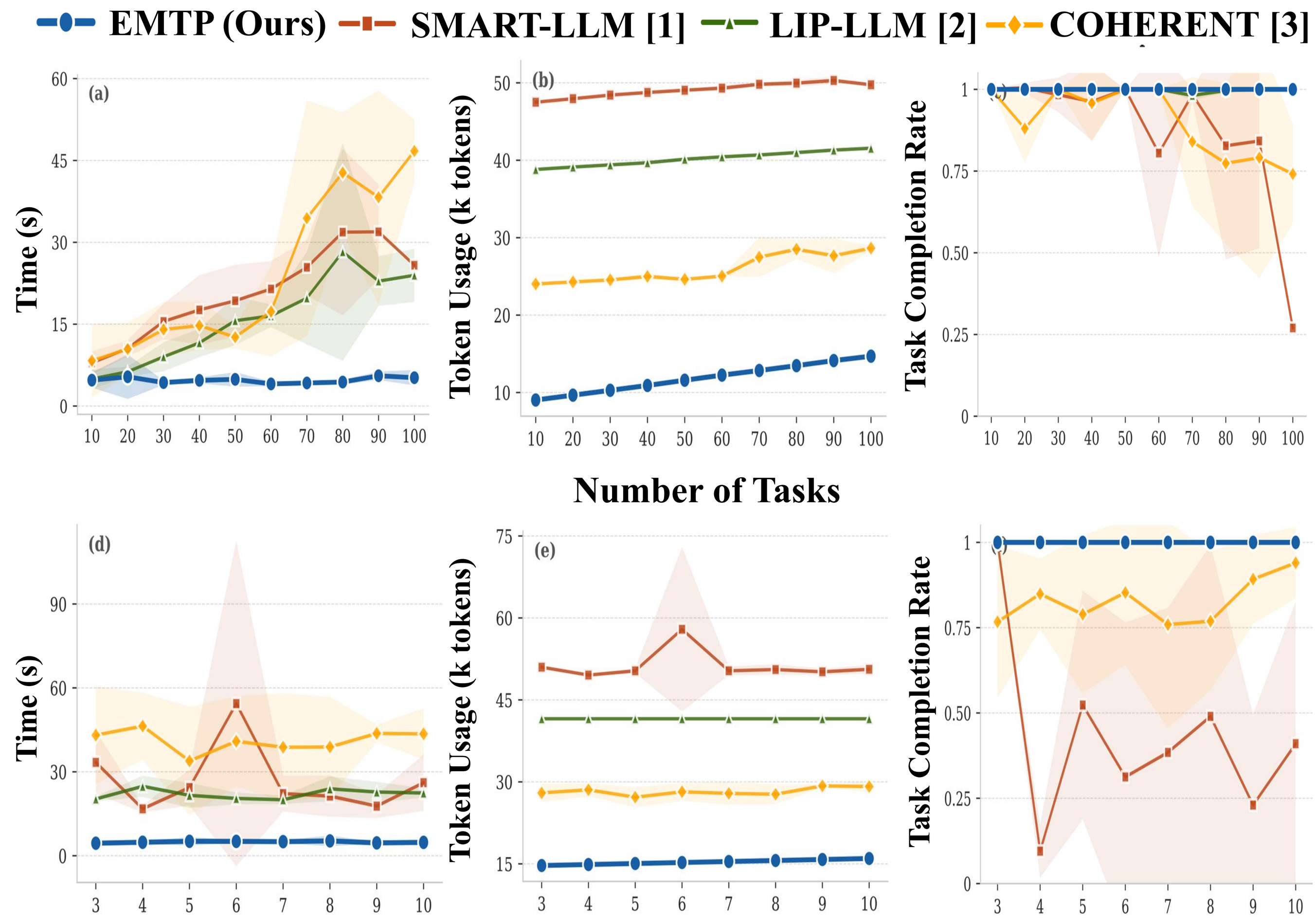
1. Initial Planning – Env. A

Const.	Method	SR ↑	TCR ↑	CSR ↑	Time ↓ (s)	Token ↓ (k)
C1	EMTP	1.00	1.00	0.93	12.1	12.38
	SMART-LLM	1.00	0.79	0.89	42.19	32.1
	LIP-LLM	1.00	0.99	0.91	37.93	36.9
	COHERENT	1.00	0.89	0.92	23.47	25.5
C2	EMTP	1.00	1.00	0.91	8.43	12.2
	SMART-LLM	0.33	0.68	0.61	59.91	30.1
	LIP-LLM	0.71	0.92	0.59	114.13	42.4
	COHERENT	1.00	0.71	0.49	48.18	27.8
C3	EMTP	1.00	0.91	0.78	10.58	12.3
	SMART-LLM	0.89	0.74	0.77	38.85	32.4
	LIP-LLM	1.00	0.90	0.67	63.77	43.8
	COHERENT	0.74	0.62	0.62	31.94	28.0
C4	EMTP	1.00	1.00	1.00	8.37	12.3
	SMART-LLM	0.78	0.94	1.00	59.94	31.3
	LIP-LLM	1.00	0.70	0.67	43.77	37.0
	COHERENT	1.00	0.90	1.00	29.77	27.7

2. Online Replanning

	Env. A				Env. B			
Method	TCR ↑	CSR ↑	Time ↓ (s)	Token ↓ (k)	TCR ↑	CSR ↑	Time ↓ (s)	Token ↓ (k)
EMTP	1.00	1.00	71.2	84.3	1.00	1.00	42.1	53.4
SMART-LLM	0.82	0.92	138.5	344.7	0.81	0.92	89.3	84.7
LIP-LLM	0.90	0.90	258.4	295.6	0.58	0.58	148.3	103.3
COHERENT	0.91	0.95	203.5	412.5	0.91	0.84	147.7	110.4

3. Scalability – Env. A



Experiments

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